

AIISH – Third Visit and Requirement Gathering Report

Institution: Maharaja Institute of Technology Mysore

Date: 18 June 2026

Location: AIISH (All India Institute of Speech and Hearing), Mysore

Faculty Guide: Ms. Sahana

Technical Guide: Mr. Karthik

Purpose of the Visit

The purpose of the third visit to AIISH was to discuss the project progress with our faculty guide, Ms. Sahana, and technical guide, Mr. Karthik. The meeting focused on understanding the practical challenges faced by children with communication difficulties and gathering detailed requirements for improving the AAC (Augmentative and Alternative Communication) application.

The guides reviewed the current application design and provided suggestions to enhance usability, accessibility, customization, and overall user experience.

Discussion Summary

During the meeting, the existing application interface and functionalities were discussed in detail. Ms. Sahana and Mr. Karthik emphasized the importance of designing an AAC system that is adaptable to the unique needs of each child.

The discussion highlighted the need for flexible communication card arrangements, simplified navigation, customizable visual content, voice personalization, and accessibility-focused design elements. The guides also stressed the importance of providing parents and therapist

with greater control over the application's customization features while ensuring ease of use for children.

Several improvements were suggested to make the application more effective and user-friendly for children with speech and communication disorders.

Requirements Suggested by the Guides

1. Customizable and Movable Communication Cards

- * Communication cards should be movable within the grid layout.
- * Parents, therapists, or caregivers should be able to rearrange cards according to the individual needs of the child.
- * The card arrangement should support easy customization and personalization.

2. Flexible Menu Navigation

- * The menu should support navigation in all directions:
 - * Up
 - * Down
 - * Left
 - * Right
- * Navigation should be simple, intuitive, and suitable for children with varying motor abilities.

3. Adjustable Grid Size

- * The grid size should be configurable based on the communication requirements of the child.
- * Different grid layouts should be available to accommodate varying levels of communication ability and visual accessibility.

4. Simplified Folder Structure

- * Folder organization should be simple and easy to understand.
- * Complex folder hierarchies should be avoided to reduce cognitive load and improve navigation efficiency.

5. Touch Recognition Optimization

- * Multiple touches occurring within a short predefined time interval should be considered as a single touch.
- * This feature will reduce accidental repeated selections and improve touch interaction accuracy.

6. Voice Modulation Options

- * The application should provide voice customization options.
- * Users should be able to select between:
 - * Male Voice
 - * Female Voice
- * Voice personalization can improve user comfort and engagement.

7. Real Image Integration and Parent Customization

- * Real photographs should be used instead of simple line drawings wherever possible.
- * Parents should be able to add custom images through image upload or direct capture.
- * The system should automatically place uploaded images into appropriate categories whenever possible.
- * Manual placement and editing options should also be available.

8. Improved Accessibility of Control Buttons

- * Frequently used action buttons such as:
 - * Clear
 - * Delete
 - * Backspace
- * Should be displayed in a larger size to improve visibility and accessibility for children.

9. Dynamic Adaptation

- * The application should include dynamic adaptation features.
- * The system should learn from user interaction patterns and adapt accordingly.
- * Frequently used cards and categories may be prioritized to improve communication efficiency.

10. Child Lock Feature

- * A child lock mechanism should be implemented to prevent accidental modification of settings and communication content.
- * Administrative functions should only be accessible to parents, caregivers, or therapists.
- * PIN or password-based protection can be considered for secure access.

Outcome of the Meeting

The meeting provided valuable insights into the practical requirements of an AAC application intended for children with communication disorders. The guides identified several enhancements aimed at improving personalization, usability, accessibility, and security.

The team gained a clear understanding of the functional requirements that need to be incorporated into the next phase of development. These recommendations will serve as the foundation for refining the application design and implementing additional features.

Conclusion

The third visit to AIISH was highly productive and focused on requirement gathering and system improvement. Ms. Sahana and Mr. Karthik provided detailed feedback on the current application and suggested several enhancements to improve its effectiveness for children with communication difficulties.

The identified requirements, including movable communication cards, customizable layouts, voice modulation, real-image support, dynamic adaptation, and child-lock functionality, will be incorporated into future development iterations to create a more accessible, user-friendly, and personalized AAC solution.